SEMF-Net: A Spatial–Spectral Edge-Enhancement-Based Multistage Fusion Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*—To tackle the issue of high-resolution hyperspectral imaging systems unable to obtain spatially fine hyperspectral images (HSIs), fusion techniques for HSIs with low-resolution and multispectral images with high-resolution have been proposed and have seen rapid development. However, current fusion algorithms based on deep learning (DL) often neglect the enhancement of spatial and spectral edge information, failing to fully exploit the correlation between spatial and spectral information and lacking comprehensive extraction of global-local spatial features. To mitigate these limitations, we proposed a spatial–spectral edge-enhancement-based multistage fusion network (SEMF-Net) for hyperspectral and multispectral image fusion. In SEMF-Net, we proposed a spectral edge mirror enhancement (SEME) block and a spatial edge enhancement (SEE) block to recover and enhance the spatial and spectral edge information during image fusion. Second, to leverage the correlation of the spatial–spectral information, we constructed the partition Transformer (Ptnformer) to realize the long-distance dependence modeling of the spectral information subregionally, which reduces the interference of unfavorable information. Finally, we designed a dual-spatial awareness (DSA) block, which can not only perform global-local perception of spatial information but also perform sub-regional perception at the same time, thus realizing more comprehensive spatial information extraction. The experimental results on four remote sensing datasets showed that SEMF-Net can alleviate the spatial and spectral edge errors in the image fusion process and obtain fusion results superior to those of the current state-of-the-art fusion algorithms. The related code is available at https://github.com/cvmdsp/SEMF-Net**

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***Index Terms*—Hyperspectral image (HSI), image fusion, multispectral image, partition Transformer (Ptnformer), spatial–spectral edge-enhancement.**

I. Introduction

A HYPERSPECTRAL image (HSI) is a 3-D data cube with rich spectral information, which covers 2-D spatial information along dense spectral channels. Due to the dense spectral bands, HSI is widely used in target detection [1], [2], precision agriculture [3], environmental monitoring [4], HSI classification [5], [6], [7], and other fields. However, during hyperspectral imaging, the spatial and spectral resolutions of HSI cannot be enhanced simultaneously due to the tradeoff between hardware resources and data sampling, so its spatial resolution is sacrificed, while its spectral resolution is enhanced [8]. The spatial resolution of low-resolution HSIs (LR-HSIs) limits the performance of downstream tasks. High-resolution multispectral images (HR-MSIs), on the other hand, happen to have the opposite properties of LR-HSI and can complement LR-HSI, so the fusion of LR-HSI with HR-MSI is a reliable method for generating high-resolution HSI (HR-HSI) [9], which is termed the hyperspectral image fusion (HIF) method.

Currently, traditional HIF methods include detail injection methods and model optimization methods. Among them, the process of detail injection involves transferring high-frequency spatial details from high-resolution images into low-resolution ones, which can be refined into component substitution (CS) methods [10], [11] and multiresolution analysis (MRA) methods [12], [13]. The CS technique utilizes the high-resolution spatial details of an HR-MSI and injects them into the LR-HSI to reconstruct an HR-HSI. The MRA method refers to the extraction of spatial details of HR-MSI through multiresolution decomposition and injecting it into LR-HSI to finally obtain HR-HSI. Detail injection methods can significantly improve the spatial detail representation of the fusion results, but they are prone to spectral shifts and introduce noise in the process of injecting details. Model optimization methods take the objective function as the core to construct mathematical models and optimization solution algorithms to find

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the optimal fusion solution, which can be specifically classified as Bayesian methods [14], [15], matrix decomposition methods [16], [17], [18], and tensor representation methods [19], [20]. Bayesian methods refer to the transformation of a fusion problem into an optimization problem with an objective function based on the Bayesian theorem and prior information. The matrix decomposition method involves unfolding the 3-D data into a 2-D matrix and then decomposing the HR-HSI into a product of the end-element matrix and the abundance matrix based on spectral unmixing or into a product of the sparse coefficients and the dictionary based on the sparse representation. The tensor representation method treats the HSI as a third-order tensor and then decomposes the HSI into factor matrices in different directions according to different decomposition methods before fusion and reconstruction. The model optimization method, which can be shown to constrain the spatial–spectral consistency, is sensitive to parameters and has poor generalization ability.

Deep learning (DL), with its powerful data modeling capability to construct nonlinear relationships among data, provides new ideas for HIF [21], [22], [23], [24], [25]. When performing HIF, the aggregation of LR-HSI and HR-MSI features requires the unification of their spatial dimension sizes, and DL-based HIF can be broadly classified into two categories according to the aggregation method. The first one involves aggregation followed by feature extraction [26], [27], [28], [29], [30], [31]. In particular, up-sampling the LR-HSI to match the spatial size of the HR-MSI, and then aggregating the up-sampled LR-HSI and the HR-MSI by cascading and other operations, and then performing feature extraction and refinement on the aggregated features. The second is the feature extraction and aggregation step-by-step [32], [33], [34], [35], [36], i.e., the spatial dimension up-sampling of LR-HSI is not done all at once, but in combination with the down-sampled HR-MSI, the features are gradually aggregated and refined at different scales, and the HR-HSI is finally obtained.

Existing DL-based HIF methods have achieved advanced fusion performance but still have some shortcomings. First, spectral and spatial features are heterogeneous, and many fusion algorithms have been designed to target feature extraction modules for spatial and spectral features, respectively, which enable the effective extraction of both spatial and spectral features from the images. However, based on the idea of the algorithm design and fusion results, most of the existing HIF algorithms ignore the importance of spatial edges and spectral edges (spectral bands located at the two ends) and are, thus, more prone to producing edge errors. Fig. 1 shows the spatial error map and spectral curve error map of the fusion results. As illustrated in Fig. 1, the spatial and spectral edges have more obvious errors, while the proposed method mitigates the generation of these two types of errors. Notably, to our knowledge, in this article, we present the concept of spectral edges for the first time in the field of HIF. Spatial edge information, as well as subtle spectral variations, have a significant impact on the application of HR-HSI in downstream missions, so it is of practical interest to explore spatial and spectral edge enhancement in HIF missions. Second, although the spatial and spectral information in HSI have different

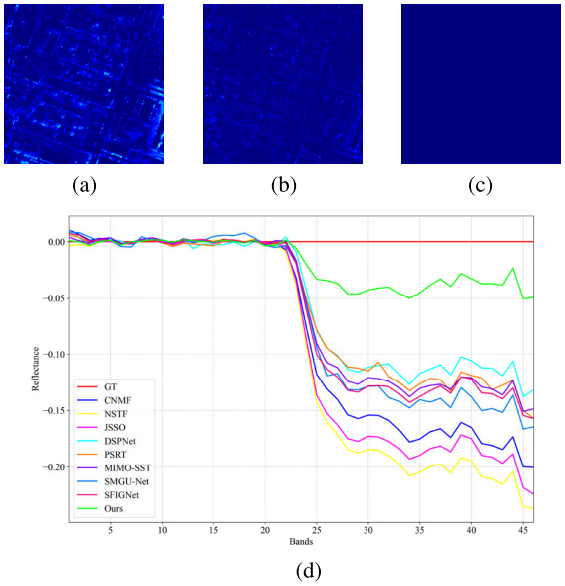


Fig. 1. Spatial and spectral error maps. The spatial error maps for SFIGNet, the method described in this article, and GT are represented by (a), (b), and (c), respectively, while (d) represents the spectral error map.

attributes, HSIs in the same scene are strongly correlated, and regions with similar spatial features tend to have more similar spectral features, but existing algorithms tend to ignore this correlation. In addition, many spatial attention-based methods perform attention without integrating global and local spatial attention, and focusing more on one or the other may lead to incomplete extraction of spatial information from the model, and the extraction of both global semantic and spatial detail information may affect the quality of the fusion results.

To overcome these limitations, in this article, we propose a spatial–spectral edge-enhancement-based multistage fusion network (SEMF-Net), which injects edge information of spatial and spectral during the fusion and is able to reduce the distortion of edge information. Simultaneously, a multistage feature extraction-fusion strategy is adopted, which considers the spatial–spectral correlation during the multistage feature extraction and is able to extract the spatial and spectral details more accurately. Moreover, a dual-spatial perception mechanism is adopted, which is able to comprehensively perceive the global semantic context and spatial details. In particular, in this article, the spectral edge mirror enhancement (SEME) block is designed to mitigate the errors at the spectral edges by adding neighboring auxiliary bands to the edge spectral bands through channel flipping and pointwise convolution. For spatial edge features, we extract spatial edge-enhanced feature maps through the proposed spatial edge enhancement (SEE) block and inject them into the spatial features of each stage to mitigate spatial edge errors in the fusion results. Each fusion stage of SEMF-Net can inherit the fusion features of the previous fusion stage and aggregate the spatial and spectral features of the current stage. In particular, a partition Transformer (Ptnformer), a dual-spatial awareness (DSA) block, and a deep-shallow attention interaction fusion (DAIF) block are designed for each fusion stage. Where Ptnformer calculates the partitioning of spectra based on spectral similarity,

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thus enhancing the co-optimization of similar spectral regions and reducing the information interference in regions with large differences. The DSA block provides comprehensive spatial awareness of the input image as a whole through the global-local spatial awareness (GLSA) block, which not only partitions the spatial region for feature extraction based on spectral similarity but also uses the image's spatial global semantics and local detail information. The DAIF block correlates and integrates the current stage of the space-spectral features with the fusion features of the previous stages. This design not only retains the uniqueness of the current stage of the spatial–spectral features but also inherits the accumulated fusion features of the previous stages and realizes the cross-modal feature integration and the inheritance of historical features. The contributions of this work can be summarized as follows.

1) Aiming at the challenges of spectral-SEE and comprehensive spatial perception in HIF, we propose the SEMF-Net, which is designed to enhance spatial and spectral edge information through a multistage fusion mechanism that integrates both cross-modal spectral–spatial features and historical fusion information from previous stages.
2) Aiming at the problem that the spatial–spectral edge information is prone to errors, we propose the concept of spectral edges for the first time and construct the SEME block and SEE block, respectively, to make up for the loss of the edge information by enhancing the spatial–spectral edge in the fusion process.
3) For spatial information extraction, we construct a DSA block, which not only can partition feature extraction of space based on spectral information but also can comprehensively perceive global spatial semantic information and local spatial details. For spectral feature extraction, we propose Ptnformer, which employs a partitioning strategy to model long-distance dependence on spectra along the spectral dimension in subregions, which can effectively alleviate the spectral distortion resulting from the large differences in spectral dimension.
4) To deeply correlate the fused features, we propose the DAIF block, which is capable of realizing multimodal information integration and feature inheritance. DAIF extracts the attention weights from different feature subspaces so as to enhance the completeness of the fused features and promote the overall performance.

II. Related Works

HIF algorithms are often categorized into traditional and DL-based algorithms, which will be described separately below. In the traditional HIF algorithms, Shahdoosti and Ghassemian [10] proposed a new spatial-domain consistent data transformation method combining the optimal filter, spatial, and spectral principal component analysis, which retains more spectral and spatial information for the fusion result. Liu et al. [16] introduced a truncated matrix decomposition method, which estimates the spatial matrix using two models, extracts the initial spectral matrix from LR-HSI, and constructs the fusion result by combining the spatial and spectral matrices. The experimental validation confirms the efficacy of this method. Long et al. [37] utilized nonlocal spatial similarity and singular value decomposition as a priori information to achieve the initial fusion and then achieved joint spatial–spectral optimization (JSSO) through a coupled optimization framework. Li et al. [38] introduced an HIF algorithm based on coupled tensor decomposition to model high spatial–spectral correlations through a regularizer and converted dictionary and core tensor estimation into a coupled tensor decomposition problem, which yielded excellent fusion results. Xu et al. [39] introduced a new two-factor decomposition model of the coupled tensor and devised a neighborhood-based alternating minimization algorithm to solve the model and obtained efficient fusion results. Although the traditional algorithm obtains a favorable fusion effect, it relies on manually setting a priori knowledge and an objective function, which limits its ability to cope with scene changes, and it is highly susceptible to noise, which easily results in a reduction in the signal-to-noise ratio of the fused result.

With the rapid advancement of DL, it has gained wide application within the domain of HIF. For instance, Deng et al. [40] designed a mixing–remixing module to realize the information interaction between global image blocks and proposed a new pyramid mixing–remixing Transformer algorithm by combining with Transformer [41] to realize HIF, which obtained good fusion results. Luo et al. [42] considered that noise is inevitably introduced during HSI acquisition and proposed a hierarchically conditioned diffusion model for HSI reconstruction. By initializing pure Gaussian noise and refining it through network iterations, this approach achieves high-quality reconstruction. Sun et al. [43] designed a spatial pyramid block and then introduced an HIF algorithm utilizing a dual-spatial–spectral pyramid network (DSPNet) integrated with Transformer. This method effectively mitigates the loss of both spatial and spectral information in the fused results. Fang et al. [44] introduced an HIF algorithm based on a spatial–spectral Transformer with a multi-input–multi-output, which obtained advanced fusion results by refining the high-frequency details through high-frequency loss of wavelets. Liu et al. [45] designed a gradual spectral mapping model, which reduces the spectral distortions by fitting the spectral information on a scale-by-scale basis. In addition, the model promotes the network to concentrate on the regions with significant spectral differences between LR-HSI and HSI through adaptive defect optimization loss, which ultimately achieves excellent fusion performance. Yu et al. [46] proposed a Transformer gate interactive U-Net-based HIF algorithm, which fully considers cross-modal information extraction and interaction and achieves high spatial and spectral information fidelity. Zhang et al. [47] explored an HIF algorithm based on a cross-frequency domain enhanced spatial–spectral fusion network, which realized the effective integration of information between spatial and frequency domains. They also designed a loss function based on the frequency domain and finally obtained excellent fusion results.

Existing DL-based HIF algorithms have made significant progress, but there is still considerable scope for further research. For example, previous methods rarely overlook the

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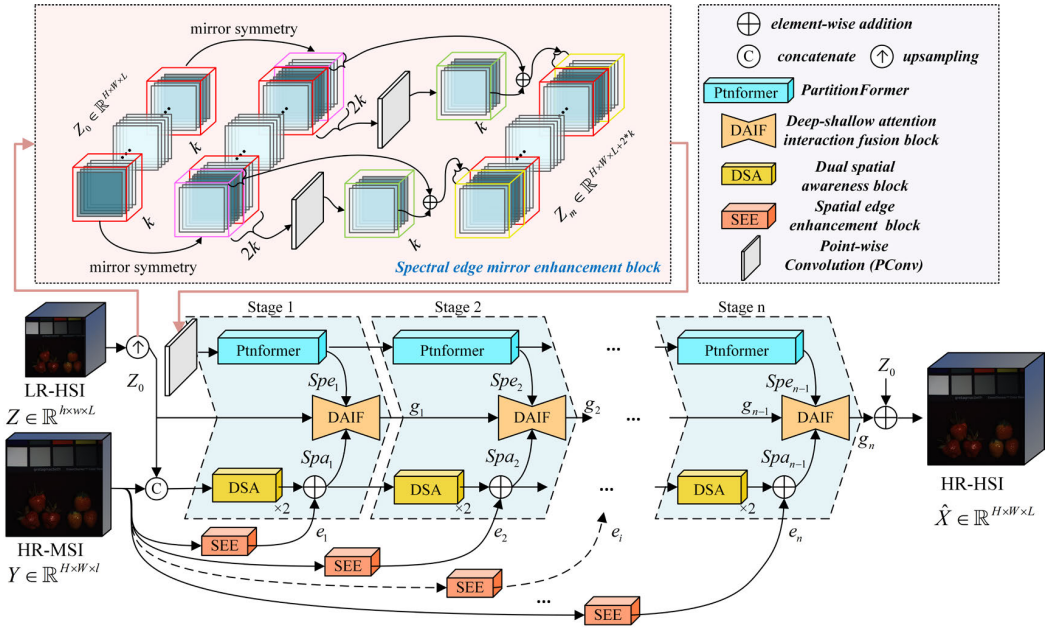


Fig. 2. Overall structure of SEMF-Net, which consists of an SEME block, SEE blocks, and multistage fusion.

loss of spatial and spectral edges throughout the fusion process, which leads to fusion results that are more likely to exhibit pronounced errors at the edge locations. In this work, we tackle this limitation by proposing the SEME and SEE blocks, which realize the enhancement of spectral edges in the front section of the model and inject spatial edge information adapted to the current stage in each stage of fusion. In addition, methods such as DSPNet design feature extraction modules for spatial and spectral information, respectively, without utilizing the self-similarity of spectral information and the correlation between spatial–spectral information. To address this problem, Ptnformer and DSA blocks are designed in this work to better explore the correlation between spatial–spectral information. Meanwhile, the DSA block also addresses the challenge of uneven extraction of global and local spatial features. Cross-modal information interaction and fusion is also an issue to be further explored, and this article designs the DAIF block to address this issue, which can effectively interactively fuse cross-modal information through the deep-shallow attention mechanism and inherit the fusion features accumulated in history. Through spatial and spectral edge enhancement, multistage feature extraction, and cross-modal feature fusion, the proposed method can effectively realize fine spatial–spectral feature fusion.

III. Method

In this section, we first describe the overall network structure of SEMF-Net. Then, we describe each module of SEMF-Net in detail, such as the SEME block, SEE block, DSA block, Ptnformer, and DAIF block.

A. Overall Network Structure

In Fig. 2, we depict the overall structure of the proposed SEMF-Net. As can be seen from Fig. 2, SEMF-Net consists of an SEME block, SEE blocks, and multistage fusion (Stage 1, Stage 2,..., Stage n). Each fusion stage contains a Ptnformer for spectral feature extraction, a DSA block for spatial feature extraction, and a DAIF block for fusing spatial–spectral features. LR-HSI, HR-MSI, and HR-HSI are denoted as $Z \in \mathbb{R}^{w\times h\times L}$, $Y \in \mathbb{R}^{W\times H\times l}$, and $X \in \mathbb{R}^{W\times H\times L}$, respectively. Where W, H, and L denote the width, height, and band number of HR-HSI, respectively. $w = W/r$, $h = H/r$, $l \ll L$, r denotes the spatial sampling rate.

SEMF-Net first up-samples the LR-HSI to obtain the feature map $Z_0 \in \mathbb{R}^{W\times H\times L}$, making it consistent with HR-MSI in the spatial dimension. Considering that when extracting the spectral feature of LR-HSI, its foremost and last bands (i.e., spectral edges) lack the assistance of the information of the neighboring bands and are prone to spectral aberrations in the fusion process. Therefore, SEMF-Net inputs Z_0 into the SEME block to enhance its spectral edges and acquire the spectral edge-enhanced feature map Z_m. Subsequently, its spectral information is integrated by pointwise convolution while retaining its complete spatial information, and the whole process can be expressed as

$$Z_m = \mathrm{PConv}\left(\mathrm{SEME}\left(Z_0\right)\right) \tag{1}$$

where SEME$(\cdot)$ represents the mapping function of the SEME block, and PConv$(\cdot)$ denotes pointwise convolution. SEMF-Net not only focuses on the enhancement of spectral edge information but also enhances spatial edge information using the SEE blocks. In order not to introduce excessive noise interference, n SEE blocks with nonshared parameters take Y as input and adaptively extract more accurate spatial edge information for different fusion stages, that is,

$$e_i = \mathrm{SEE}_i\left(Y\right), \quad i = 1, 2, \ldots, n \tag{2}$$

where SEE$_i(\cdot)$ indicates the mapping function of the ith SEE block, and e_i indicates the output of the ith SEE block. Subsequently, edge information e_i is injected into the spatial features of each stage to enhance the spatial edge information.

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SEMF-Net consists of n fusion stages. The input for the first fusion stage is Z_m, Z_0, and $[Z_0, Y]$, where $[\cdot,\cdot]$ indicates a cascading operation. In particular, the Z_m inputs into Ptnformer for extracting the feature map Spe_1 with rich spectral information and the $[Z_0, Y]$ inputs into two cascaded DSA blocks to extract the feature map Spa_1 with complete spatial information. Then Spe_1, Spa_1, and Z_0 are input into the DAIF block to complete the first stage of spatial–spectral information extraction and fusion. The process can be defined as follows:

$$\begin{cases} \text{Spa}_1 = \text{DAS}_1([Z_0, Y]) + \text{SEE}_1(Y) \\ \text{Spe}_1 = \text{Ptnformer}_1(Z_m) \\ g_1 = \text{DAIF}_1(\text{Spa}_1, \text{Spe}_1, Z_0) \end{cases} \tag{3}$$

where $\text{DAS}_1(\cdot)$, $\text{Ptnformer}_1(\cdot)$, and $\text{DAIF}_1(\cdot)$ denote the mapping functions corresponding to each of the two cascaded DSA blocks, the Ptnformer block, and the DAIF block in the first stage, and g_1 represents the fusion feature map of the first stage. The output features Spe_1, Spa_1, and g_1 of the first stage are used as inputs to the second fusion stage for further spatial–spectral feature extraction and fusion, and so on up to the nth fusion stage. Then the 2nd–nth fusion stage can be expressed as

$$\begin{cases} \text{Spa}_i = \text{DAS}_i(\text{Spa}_{i-1}) + \text{SEE}_i(Y) \\ \text{Spe}_i = \text{Ptnformer}_i(\text{Spe}_{i-1}), \\ g_i = \text{DAIF}_i(\text{Spa}_i, \text{Spe}_i, g_{i-1}) \end{cases} \quad i = 2, \ldots, n \tag{4}$$

where $\text{DAS}_i(\cdot)$, $\text{Ptnformer}_i(\cdot)$, and $\text{DAIF}_i(\cdot)$ denote the mapping functions corresponding to each of the cascaded DSA block, Ptnformer block, and DAIF block in the ith stage. Spe_i, Spa_i, and g_i denote the spectral, spatial, and fusion feature maps of the ith stage. After n stages of fusion, SEMF-Net obtained a fusion feature map g_n with fine spectral features, spatial edges, and texture features. To mitigate the loss of spatial and spectral information from the source images, long jump connections are also incorporated in SEMF-Net to obtain the final fusion results $\hat{X}$, that is,

$$\hat{X} = g_n + Z_0. \tag{5}$$

In this article, the proposed SEME block, SEE block, Ptnformer, DSA block, and DAIF block are described in detail.

B. Spatial and Spectral Edge Enhancement

The spatial and spectral dimensions comprise the core information structure of the HSI data cube. In the spatial dimension, the edge information is defined as the region where the gradient value changes sharply, which is important for target detection and classification tasks. In the spectral dimension, we define the bands located at the head and tail of the HSI as spectral edges. The spectral information of the HSI varies continuously, with a certain degree of similarity between the spectral bands, and bands in the middle are complementary and assisted by the front and back bands, which can realize finer fusion of spectral information. The head and tail bands are prone to serious spectral distortion due to the lack of front and back band information. The subtle differences in spectral information may affect the discrimination of substances, so it is necessary to fuse the spectral information completely.

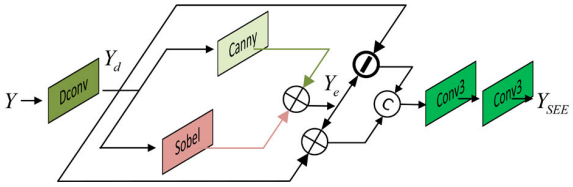


Fig. 3. Network structure of SEE block, which consists of Sobel operator, Canny operator, and convolutional layer.

Existing HIF algorithms often lack attention to spatial–spectral edges, resulting in spectral edge distortion or unclear spatial edges in the fusion results. In this article, the SEE block is designed for enhancing the spatial edge information, and the SEME block is designed for enhancing the learning of spectral edge information in order to realize the fine fusion of spectral information. This article will introduce the SEME block and the SEE block, respectively.

1) SEME Block: The SEME block is shown in the pink area in Fig. 2, and it mainly targets the spectral edge bands for enhancement. In particular, first mirror-reverse the k bands at the head and tail, and concatenate them to the head and tail of the image, respectively. This operation ensures the continuity and similarity of spectral information, which is defined as

$$Z_1 = \left[\mathscr{M}\left(Z_0^{1\sim k}, Z_0, \mathscr{M}\left(Z_0^{L-k-1\sim L}\right)\right]\right. \tag{6}$$

where $Z_0^{1\sim k}$ indicates the k bands located at the head of Z_0, $Z_0^{L-k-1\sim L}$ indicates the k bands located at the tail of, and $\mathscr{M}(\cdot)$ indicates a mirror flip. $[\cdot,\cdot]$ indicates cascading along the channel. Then, the first $2k$ channels and the last $2k$ channels of Z_1 are input into pointwise convolution for channel-to-channel reorganization to learn the enhanced spectral edge features. In addition, to improve the learning ability of the SEME block, we also incorporate the short jump connection, which can be formulated as follows:

$$Z_m^{1\sim k} = \text{PConv}\left(Z_1^{1\sim 2k}\right) + \mathscr{M}\left(Z_1^{1\sim k}\right) \tag{7}$$

$$Z_m^{L-k-1\sim L} = \text{PConv}\left(Z_1^{L-2k-1\sim L}\right) + \mathscr{M}\left(Z_1^{L-k-1\sim L}\right) \tag{8}$$

where $\text{PConv}(\cdot)$ denotes pointwise convolution. The output of the SEME block is acquired through the splicing operation, that is,

$$Z_m = \left[Z_m^{1\sim k}, Z_0, Z_m^{L-k-1\sim L}\right] \in \mathbb{R}^{W\times H\times L+2k}. \tag{9}$$

2) SEE Block: Fig. 3 depicts the architectural design of the SEE block. The SEE block first changes the channel of the feature map to 1 by depthwise convolution. Then, to make the edge feature extraction adaptable to complex scenes and noisy environments, the SEE block adopts two kinds of edge operators, Canny and Sobel, to extract the feature edge information. Among them, the Sobel operator calculates the horizontal and vertical gradients. However, the edge localization obtained by the Canny operator is more accurate and noise-resistant. The two edge features are summed to get the complete edge feature Y_e. The detailed operation is outlined as follows:

$$Y_d = \text{DConv}(Y) \tag{10}$$

$$Y_e = E_{\text{Canny}}(Y) + E_{\text{Sobel}}(Y) \tag{11}$$

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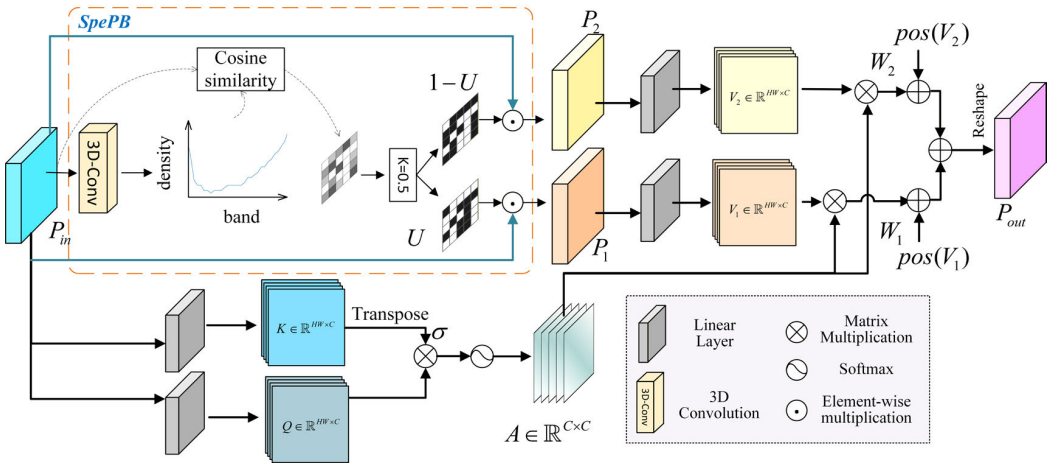


Fig. 4. Structure of PS-MSA with partitioned spectral MSA computation on feature space.

where DConv(·) represents depthwise convolution. $E_{\text{Canny}}(\cdot)$ and $E_{\text{Sobel}}(\cdot)$ denote the Canny and Sobel operators, respectively. Although the addition of the edge features Y_e and the feature map Y_d results in a feature map with enhanced edge features, the simple addition may result in weakening of the texture features of the image. Consequently, we simultaneously obtain a feature map with attenuated edges and prominent texture by subtracting Y_e and Y_d. Then, the output of the SEE block is obtained by cascading and convolution operations, that is,

$$Y_{\text{SEE}} = \text{Conv3}\left(\text{Conv3}\left(\left[Y_d + Y_e, Y_d - Y_e\right]\right)\right) \tag{12}$$

where Conv3(·) denotes a convolutional operator with a step size of 1 and a convolutional kernel size of 3×3.

C. Partition Transformer

The Transformer structures proposed in [48] and [49] can extract spectral information, but they uniformly process the input, overlooking the issue of spectral features from different regions interfering with each other. The same scene may contain regions with large spectral differences, and processing these regions simultaneously may cause mutual interference between the information. So, based on a partitioning of the feature space guided by spectral similarity, we designed the Ptnformer, building upon the Transformer. We substitute the multihead self-attention (MSA) in Transformer with partition spectralwise MSA (PS-MSA), depicted in Fig. 4.

Ptnformer first partitions input features using the spectral partition block (SpePB), then performs spectral MSA across regions with similar spectral characteristics. Each spectral vector is treated as a token to explore long-range dependencies in the spectral dimension. In particular, SpePB simulates a discriminative spectral curve through a 3-D convolution. It then calculates the cosine similarity between this spectral curve and the spectral curve at each pixel position, yielding a similarity score map. Subsequently, the similarity score map is binarized based on the threshold to obtain two partition maps. The computational process can be represented as follows. Subsequently, the similarity score map is binarized based on the threshold ($K = 0.5$) to obtain two partition maps. The computational process can be represented as follows:

$$\text{SpeV} = \text{Conv3D}\left(P_{\text{in}}\right) \tag{13}$$

$$U\left(i, j\right) = \text{Cos}\left(\text{SpeV}, P_{\text{in}}\left(i, j\right)\right) \tag{14}$$

$$\begin{cases} U\left(i, j\right) = 0, & U\left(i, j\right) < 0.5 \\ U\left(i, j\right) = 1, & U\left(i, j\right) \geq 0.5 \end{cases} \tag{15}$$

where Conv3D(·) denotes the 3-D convolution operation, SpeV denotes the spectral curve, Cos(·) denotes the cosine similarity operator, and $U(i, j)$ indicates the similarity score value at position (i, j). The P_{in} is multiplied by U and $1 - U$, respectively, to obtain $P_1 = P_{\text{in}} \cdot U$ and $P_2 = P_{\text{in}} \cdot (1 - U)$, thereby partitioning P_{in}. Subsequently, we perform dimensional reshaping of the features P_{in}, P_1, and $P_2 \in \mathbb{R}^{W \times H \times L}$, and map them to the spectral query Q, key K, and values V_1, V_2, through linear layers, that is,

$$\begin{cases} \{Q, K\} = P_{\text{in}} W^{Q}, P_{\text{in}} W^{k} \\ \{V_1, V_2\} = P_1 W^{V_1}, P_2 W^{V_2} \end{cases} \tag{16}$$

where W^Q, W^K, W^{V_1}, and W^{V_2} denote the learnable weights. Subsequently, $\{Q, K, V_1, V_2\}$ was split into N heads, such as $Q = [Q_1, Q_2, \ldots, Q_N]$, $K = [K_1, K_2, \ldots, K_N]$, $V_1 = [V_1^1, V_1^2, \ldots, V_1^N]$, and $V_2 = [V_2^1, V_2^2, \ldots, V_2^N]$. Then, we compute the self-attention map for each head as follows:

$$\begin{aligned} A_n &= \text{softmax}\left(\sigma_n K_n^{\text{T}} Q_n\right) \\ \text{head}_1^n &= A_n V_1^n, \quad \text{head}_2^n = A_n V_2^n \end{aligned} \tag{17}$$

where σ_n represents learnable parameters, K_n^{T} represents the transpose of matrix K_n, and softmax(·) indicates the softmax activation function. The final output is produced by concatenating the N heads, applying a linear projection, and adding a positional embedding

$$\begin{aligned} P_{\text{out}'} = &\left(\left[\text{head}_1^1, \text{head}_1^2, \ldots, \text{head}_1^n\right]\right) W_1 + \text{pos}\left(V_1\right) \\ &+ \left(\left[\text{head}_2^1, \text{head}_2^2, \ldots, \text{head}_2^n\right]\right) W_2 + \text{pos}\left(V_2\right) \end{aligned} \tag{18}$$

where W_1 and W_2 represent learnable parameters, while pos(·) is used to generate position embeddings. pos(·) is composed of a 3×3 convolution and a GELU activation function to

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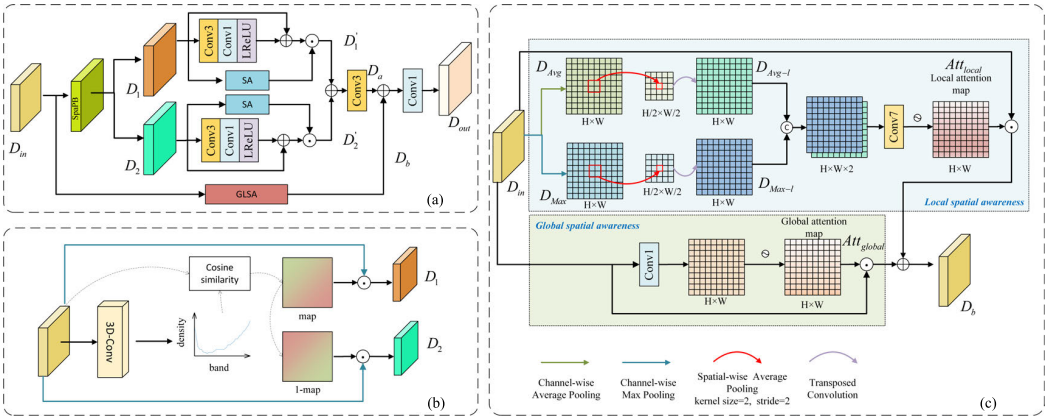


Fig. 5. Structure of the DSA block, which consists of partitioned spatial awareness and GLSA. (a) DSA. (b) SpaPB. (c) GLSA.

encode the positional information of different spectral bands. This process can be represented as

$$\text{pos}(V) = \text{GELU}(\text{Conv3}(V)) \tag{19}$$

where Conv3(·) is the convolution operation, GELU(·) is a GELU activation function. Finally, a reshape operation is performed on $P_{\text{out}'}$ to obtain the output $P_{\text{out}} \in \mathbb{R}^{H\times W\times C}$.

D. Dual-Spatial Awareness Block

To achieve a holistic perception of the global spatial semantics and local spatial details within images, we construct the DSA block. The structural design of the DSA block is depicted in Fig. 5(a), in which the first spatial perception adopts a partitioning strategy similar to that of the Ptnformer and extracts spatial features in a sub-region by utilizing the spatial partition block (SpaPB), and the specific structure is provided in Fig. 5(b). The second spatial perception is global-local spatial perception. In this article, we extract the global-local spatial attention across the global and local scopes, respectively, from the input features by constructing a GLSA block. Fig. 5(c) illustrates the architecture of the GLSA block.

From Fig. 5(a), it can be seen that the first and second spatial perceptions of the DSA block are carried out in parallel, and in the first spatial perception, the initial feature map D_{in} is first partitioned into the feature space by the SpaPB block, and its computational process is similar to that of the SpePB process in the Ptnformer. Unlike SpePB, the similarity score maps in SpaPB are not binarized; this is because spatial features have continuity in the spatial dimension, and the direct removal of a part of the features is not conducive to the model's perception of the spatial features. SpaPB divides input feature D_{in} into D_1 and D_2. Subsequently, the first spatial perception uses convolution operations to perform preliminary spatial feature extraction on D_1 and D_2, retains the initial partition features through skip connections, and then adds spatial attention for spatial feature perception, namely,

$$D_1' = (\text{LReLU}(\text{Conv1}(\text{Conv3}(D_1))) + D_1) \cdot \text{SA}(D_1) \tag{20}$$

$$D_2' = (\text{LReLU}(\text{Conv1}(\text{Conv3}(D_2))) + D_2) \cdot \text{SA}(D_2) \tag{21}$$

where Conv1(·) and Conv3(·) denote convolution operations with kernel sizes of 1×1 and 3×3, respectively, while LReLU(·) denotes the LeakyReLU activation function. SA(·) refers to the spatial attention mapping function, which consists of max pooling, average pooling, convolution operations, and the Sigmoid, with its configuration following the convolutional block attention module (CBAM). Finally, we perform feature aggregation on D_1' and D_2' by addition and convolution operations

$$D_a = \text{Conv3}(D_1' + D_2') \tag{22}$$

where D_a is a feature map that aggregates partition features. The first layer of spatial perception extracts and enhances features from different regions, which are aggregated in D_a.

Although the features are partitioned for processing to alleviate the mutual interference between heterogeneous features to a certain extent and also achieve the auxiliary enhancement between homogeneous features, it may also cause the problem of unnatural transition between regions or nonuniform style. To address this problem, the GLSA block performs a second spatial perception, focusing on the global and local spatial features as a whole. In particular, the GLSA block is divided into local space perception and global space perception. In the local spatial perception part, the input feature maps first go through the average pooling AvgPool(·) and max pooling MaxPool(·) of channel dimensions to obtain the feature maps with average semantic and salient features, which are computed as follows:

$$\begin{cases} D_{\text{Avg}} = \text{AvgPool}(D_{\text{in}}) \\ D_{\text{Max}} = \text{MaxPool}(D_{\text{in}}). \end{cases} \tag{23}$$

Subsequently, the GLSA block performs average pooling with both kernel and stride sizes set to 2 ($\text{AvgPool}_{2,2}(\cdot)$) on D_{Avg} and D_{Max} in the spatial dimension to acquire the local spatial features of them. Then, the GLSA block performs feature reconstruction on the local feature map through transposed convolution to further extract its spatial information while restoring its spatial dimension, that is,

$$\begin{cases} D_{\text{Avg}-l} = T\text{Conv}\left(\text{AvgPool}_{2,2}\left(D_{Avg}\right)\right) \\ D_{\text{Max}-l} = T\text{Conv}\left(\text{AvgPool}_{2,2}\left(D_{\text{Max}}\right)\right) \end{cases} \tag{24}$$

where TConv(·) represents transposed convolution. Subsequently, we integrate the information from two local feature

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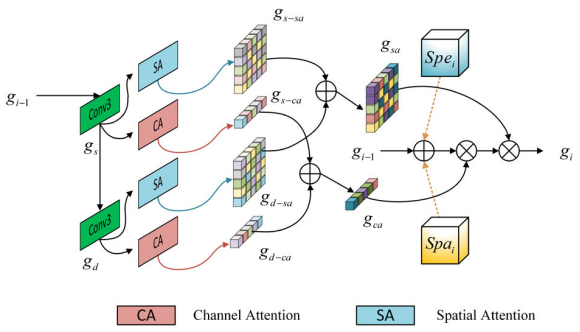


Fig. 6. Structure of the DAIF block, which realizes feature interactions through deep and shallow attentions.

maps $D_{\text{Avg}-l}$ and $D_{\text{Max}-l}$, and generate local attention weights, that is,

$$\text{Att}_{\text{local}} = \text{Sigmoid}\left(\text{Conv7}\left(\left[D_{\text{Avg}-l}, D_{\text{Max}-l}\right]\right)\right) \tag{25}$$

where Conv7(·) represents a convolution layer with a kernel size of 7×7, and Sigmoid(·) represents a Sigmoid activation function.

In the global spatial perception part, the input features are compressed to one channel by a 1×1 convolution to integrate the band information, which realizes the global attention over the spatial domain. In addition, the LeakyReLU activation function is added to enhance the expressive ability of the GLSA block. The computational process of generating global attention weights can be expressed as follows:

$$\text{Att}_{\text{global}} = \text{Sigmoid}\left(\text{LReLU}\left(\text{Conv1}\left(D_{\text{in}}\right)\right)\right). \tag{26}$$

Subsequently, the GLSA block applies global and local attention to D_{in}, aggregating the global and local spatial information to form the final output D_b. The calculation procedure is defined as follows:

$$D_b = \text{Att}_{\text{local}} \cdot D_{\text{in}} + \text{Att}_{\text{global}} \cdot D_{\text{in}}. \tag{27}$$

Ultimately, the DSA block aggregates the outputs of the first and second weight spatial sensing to obtain the output of the DSA block, that is,

$$D_{\text{out}} = \text{Conv1}\left(D_a + D_b\right). \tag{28}$$

E. Deep-Shallow Attention Interaction Fusion

The Ptnformer and the DSA block extract the spectral and spatial features, respectively, while the DAIF block interactively fuses the spectral–spatial features of the current stage and the fused features of the previous stages, and its detailed structure is illustrated in Fig. 6. The DAIF block employs the simple spatial and channel attention in CBAM and cleverly realizes the interactive fusion of deep and shallow attention. Shallow attention focuses on smoother spatial and spectral features, while deep attention focuses on spatial and spectral features with more pronounced gradient changes.

In particular, the DAIF block first performs shallow and deep feature extraction on the fused features g_{i-1} from the previous stage, that is,

$$g_s = \text{Conv3}\left(g_{i-1}\right), \quad g_d = \text{Conv3}\left(g_s\right) \tag{29}$$

where g_s represents shallow features and g_d represents deep features. Subsequently, the DAIF block calculates the attention maps of deep and shallow features through g_s and g_d, respectively, that is,

$$g_{s-\text{sa}} = \text{SA}\left(g_s\right), \quad g_{s-\text{ca}} = \text{CA}\left(g_s\right) \tag{30}$$

$$g_{d-\text{sa}} = \text{SA}\left(g_d\right), \quad g_{d-\text{ca}} = \text{CA}\left(g_d\right) \tag{31}$$

where SA(·) and CA(·) denote the spatial and channel attention operators, respectively. Next, we combine deep and shallow attention weights to obtain attention weights that comprehensively focus on deep and shallow features, and use them to perform fusion interactions on the input feature maps, that is,

$$\begin{cases} g_{\text{sa}} = g_{s-\text{sa}} + g_{d-\text{sa}} \\ g_{\text{ca}} = g_{s-\text{ca}} + g_{d-\text{ca}} \\ g_i = \left(\text{Spa}_i + \text{Spe}_i + g_{i-1}\right) \cdot g_{\text{ca}} \cdot g_{\text{sa}} \end{cases} \tag{32}$$

where g_i is the fusion feature of the current stage, which aggregates the spatial and spectral features of the current stage while inheriting the fusion features of the previous stage. It can be seen that DAIF retains the fusion information of the historical stages, ensures that the spatial–spectral information is continuously accumulated and optimized, and mitigates the loss of it.

F. Loss Function

Since the SEMF-Net fully considers the spatial edges, spectral edges, and texture information of images, only a simple L_1 loss is selected to train the SEMF-Net model, which can be represented by

$$\text{Loss} = \frac{1}{\text{HWL}} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{k=1}^{L} \left\| X_{i,j,k} - \hat{X}_{i,j,k} \right\|_1 \tag{33}$$

where i, j, and k represent the pixel positions, $\hat{X}$ represents the fusion image, and X represents the ground truth (GT). W, H, and L represent the width, height, and number of bands of $\hat{X}$ and X, respectively, and $\|\cdot\|$ represents the L_1-norm.

IV. Experimental Results and Analysis

To evaluate the efficacy of the SEMF-Net, comparative experiments with the state-of-the-art HIF algorithms are performed on three simulated and one real-world HSI dataset. The three simulated HSI datasets are the Harvard dataset [50], the Houston dataset [51], and the Pavia University (PaviaU) dataset, and the real HSI dataset is the WorldView-3 (WV3) dataset [52]. In addition, we validate the effectiveness of the proposed blocks through ablation studies.

A. Datasets

The Harvard dataset comprises 50 HR-HSIs capturing both indoor and outdoor scenes, each with a size of $1024 \times 1024 \times 31$, where 31 bands uniformly span the wavelength range from 420 to 720 nm, with a sampling step of 10 nm. The Harvard dataset was randomly partitioned into a training set of 35 images and a testing set comprising the remaining samples. The Houston dataset, provided by the HSI Analytical

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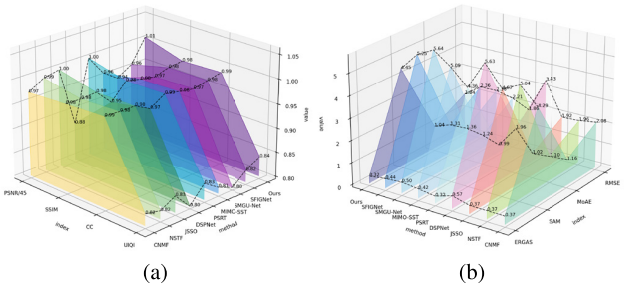
Laboratory at the University of Houston and the National Center for Airborne Laser Mapping, has an image size of $1202 \times 4172 \times 48$, with 48 bands covering the range of 380–1050 nm with a sampling step of 10 nm, and 46 bands are retained after removal of noise and absorption bands. We divide HR-HSI into 14 regions and crop them into $512 \times 512 \times 46$ blocks. Finally, seven image blocks are allocated to the training and testing sets, respectively. The PaviaU dataset is an HSI taken over Pavia, northern Italy, with a spatial size of 610×340, a band number of 103, and a geometric resolution of 1.3 m. We crop out a 256×256 region for testing, and the remaining regions are used for training. The WV3 dataset contains the panchromatic image (PAN) and MSI captured by the WV3 satellite, with a spatial resolution of 0.3 m for PAN and 1.2 m for MSI with eight bands. In this article, the WV3 dataset provided by Deng et al. [52] is used for training and testing. The training set contains 9000 pairs of PAN and MSI image blocks with sizes of $64 \times 64 \times 1$ and $16 \times 16 \times 8$, respectively. The testing set contains 20 pairs of PAN and MSI with sizes of $512 \times 512 \times 1$ and $128 \times 128 \times 8$, respectively.

For the simulated dataset, HR-HSI is spatially downsampled and spectrally downsampled to obtain LR-HSI and HR-MSI, respectively, according to Wald's protocol. In particular, in this article, HR-HSI is downsampled by 32-fold bilinear interpolation and fuzzy-processed by a Gaussian fuzzy kernel of size 5×5 and standard deviation of 2 to obtain LR-HSI. HR-MSI is obtained by applying a spectral response function to HR-HSI. This article employs distinct spectral response functions tailored to each dataset: the Nikon D700 camera for the Harvard dataset, WorldView-2 for the Houston dataset, and IKONOS-2 for the PaviaU dataset.

B. Experimental Details

To assess the performance of the SEMF-Net, it is compared with eight state-of-the-art algorithms, including three traditional algorithms: the nonlocal sparse tensor factorization (NSTF) fusion algorithm [19], the coupled nonnegative matrix factorization (CNMF) unmixing fusion algorithm [17], and the JSSO fusion algorithm [37], and five DL-based methods: the DSPNet [43], the fusion algorithm based on the pyramid shuffle-and-reshuffle Transformer (PSRT) [40], the fusion algorithm based on the multi-input–multi-output spatial–spectral Transformer (MIMO-SST) [44], the fusion algorithm based on the the spatial–spectral unfolding network (SMGU-Net) with mutual guidance [53], and the fusion algorithm based on the spatial-frequency dual-domain implicit guided sampling network (SFIGNet) [22].

To objectively verify the effectiveness of the SEMF-Net, eight objective evaluation indices are used to evaluate the simulated dataset: the error of relative global accuracy in synthesis (ERGAS), mean absolute error (MoAE), root-mean-square error (RMSE), spectral angle mapper (SAM), peak-signal-to-noise ratio (PSNR), structural similarity (SSIM), correlation coefficient (CC), and universal image quality index (UIQI). For the real dataset, we use the quality with no reference (QNR) index and spectral distortion index (D_λ) to evaluate the fusion results objectively. The higher the value of PSNR, the better the fusion quality of the model; the lower the value of RMSE, ERGAS, MoAE, SAM, and D_λ, the higher the quality of the fused image; and the closer the value of CC, UIQI, SSIM, and QNR is to 1, the higher the quality of the fused image.



(a) (b)

Fig. 7. Visualization of quantitative evaluation of different algorithms on the Harvard dataset. (a) PSNR, SSIM, CC, and UIQI. (b) ERGAS, SAM, MoAE, and RMSE. Larger colored areas in (a) indicate better metrics, and vice versa in (b).

TABLE I
QUANTITATIVE EVALUATION OF DIFFERENT ALGORITHMS ON THE HARVARD DATASET

Methods	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
CNMF	0.3727	1.1623	2.0848	**4.2884**	43.8417	0.9802	0.9852	0.8244
NSTF	0.3726	1.1012	1.9590	5.0449	44.6787	0.9790	0.9836	0.8170
JSSO	0.3693	1.0244	1.9210	4.6733	44.9688	0.9818	0.9836	0.8338
DSPNet	0.5710	1.9600	3.4264	5.6283	39.4631	0.9508	0.9670	0.7584
PSRT	**0.3210**	**0.9915**	1.8577	4.3597	45.1404	**0.9835**	0.9860	0.8339
MIMO-SST	0.4204	1.2392	2.2112	5.0880	43.2849	0.9763	0.9797	0.8133
SMGU-Net	0.4994	1.3576	2.3776	5.6448	42.3336	0.9711	0.9746	0.8003
SFIGNet	0.4385	1.3084	2.3633	5.2541	43.1754	0.9799	0.9805	0.8249
Ours	0.3222	1.0417	**1.8360**	4.4511	**45.3519**	**0.9835**	**0.9869**	**0.8397**

The DL framework used in this article is PyTorch 1.11.0, and the related algorithms are compiled using Python 3.10. The training and testing of this article are performed on a Windows 11 operating system equipped with an 11th Gen Intel[1] Core[2] i9-11900K at 3.50 GHz and an NVIDIA RTX A6000 GPU. In this article, the network model is optimized using the adaptive moment estimation algorithm (Adam) optimizer, where the batch size is set to 32, the epochs are set to 3000, and the learning rate is set to 0.0001. To make a fair comparison, this article used the same device and datasets to train and test the comparison algorithm based on its hyperparameter settings.

C. Experimental Results on Simulated Datasets

1) Experimental Results on the Harvard Dataset: In this article, we compare the proposed SEMF-Net with eight comparison algorithms on the Harvard dataset. Table I presents the average values of the objective evaluation metrics for each fusion algorithm, calculated across all test images. The optimal values are emphasized in bold, and the suboptimal values are indicated with an underline in Table I. Among the eight evaluation metrics, SEMF-Net obtained five optimal values and one suboptimal value. Notably, SEMF-Net surpasses the suboptimal competitor by 0.2115 dB in PSNR, indicating that its fusion results more closely approximate the GT. In addition,

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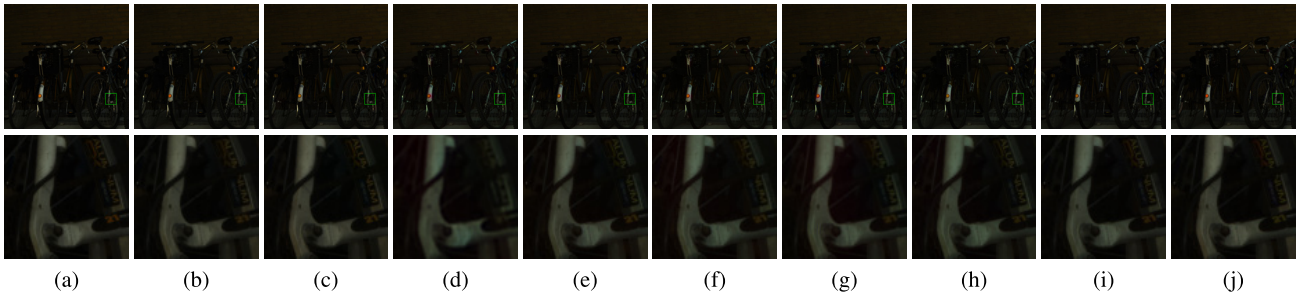


Fig. 8. Pseudo-color images (the 29th, 19th, and 9th spectral bands) of fusion results of different algorithms on the Harvard dataset. (a) CNMF, (b) NSTF, (c) JSSO, (d) DSPNet, (e) PSRT, (f) MIMO-SST, (g) SMGU-Net, (h) SFIGNet, (i) Ours, and (j) GT.

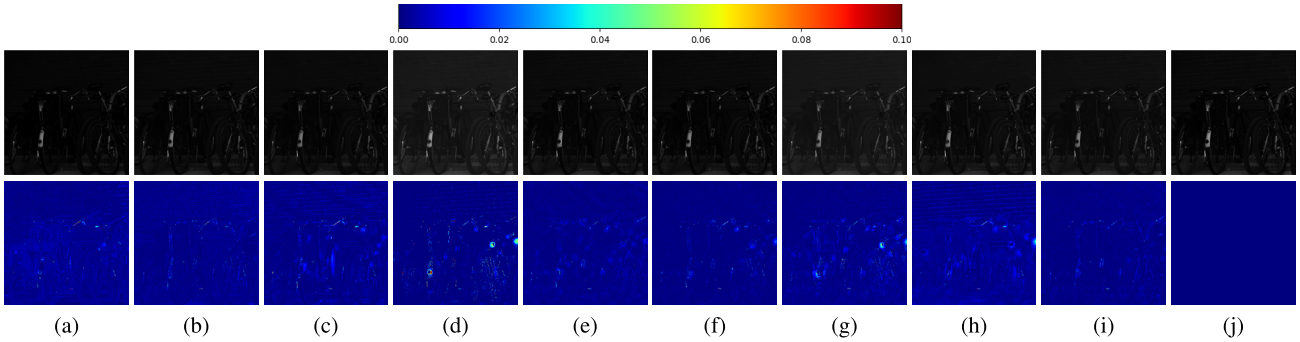


Fig. 9. Single-band grayscale and error maps (14th band) of the fusion results of different algorithms on the Harvard dataset. (a) CNMF, (b) NSTF, (c) JSSO, (d) DSPNet, (e) PSRT, (f) MIMO-SST, (g) SMGU-Net, (h) SFIGNet, (i) Ours, and (j) GT.

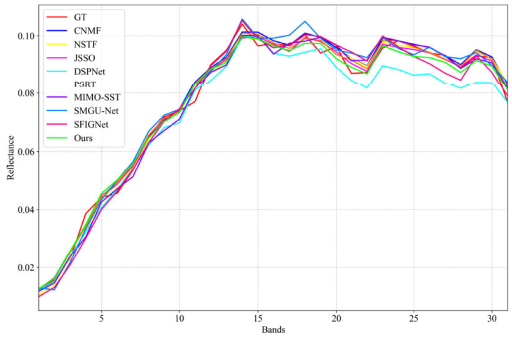


Fig. 10. Spectral curves of fusion results of different algorithms on the Harvard dataset.

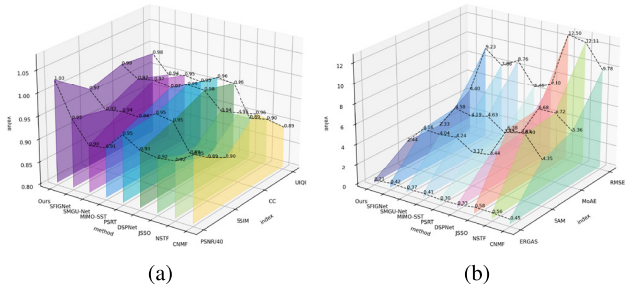


Fig. 11. Visualization of quantitative evaluation of different algorithms on the Houston dataset. (a) PSNR, SSIM, CC, and UIQI. (b) ERGAS, SAM, MoAE, and RMSE. Larger colored areas in (a) indicate better metrics, and vice versa in (b).

to visualize the objective metrics more intuitively, a waterfall diagram of them is plotted in Fig. 7. Fig. 7(a) illustrates PSNR, SSIM, CC, and UIQI, with a larger area of the waterfall plot indicating a better indicator. Fig. 7(b) demonstrates ERGAS, MoAE, RMSE, and SAM, and the smaller area of the waterfall plot indicates the better metrics. Comprehensively analyzing each evaluation metric, SEMF-Net obtains a superior fusion effect. In addition, PSRT obtains the optimal values on three metrics and is also able to effectively realize HSI image fusion.

To assess the visual effect of the fused results, an image is randomly selected from the test images for demonstration. Fig. 8 presents the pseudo-color visualization of the fused "imgf5" from the Harvard dataset, and the details in the green box are zoomed in. As evidenced by the zoomed-in area, the fusion results of DSPNet, MIMO-SST, and SFIGNet have obvious color distortion, which may be due to spectral bands shifted during the fusion process. From the number "21" in the fusion image, it is evident that the fusion results of NSTF, DSPNet, and SMGU-Net have spatial blurring problems. Overall, SEMF-Net achieves superior spatial clarity and produces color that more closely aligns with the GT. Furthermore, Fig. 9 demonstrates the grayscale and error maps of the 14th band of the fusion results, which shows that the fusion image of the SEMF-Net has the smallest overall error. It is noteworthy that the fusion image of the SEMF-Net has the smallest edge error at the brick wall, while the fused images of the other fusion algorithms have regular edge errors, which indicates that the SEE block can efficiently extract the edge information and integrate it into the spatial features. To observe the spectral characteristics of the fused results, the spectral curves at the random position of "imgf5" are plotted in Fig. 10. The fused images of DSPNet and CNMF have more obvious spectral errors in some bands, while the fused images of other algorithms can fit the GT better. Overall, SEMF-Net can better extract spatial–spectral information and reduce the distortion of details in the cross-modal fusion process.

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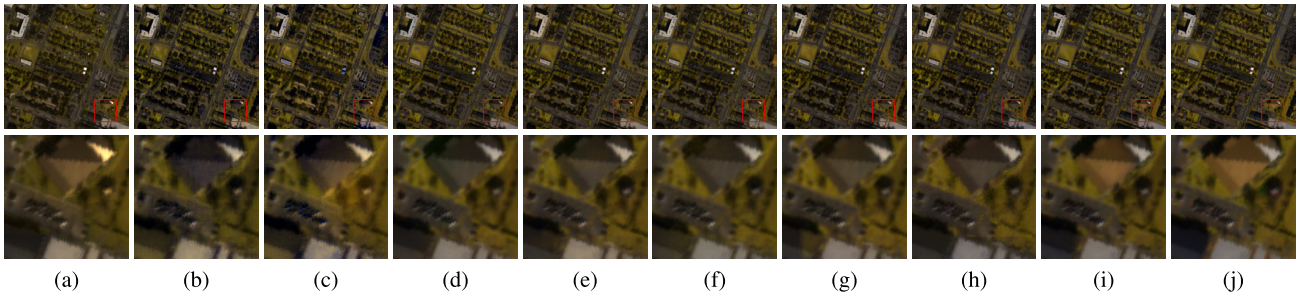


Fig. 12. Pseudo-color images (the 42th, 30th, and 15th spectral bands) of fusion results of different algorithms on the Houston dataset. (a) CNMF, (b) NSTF, (c) JSSO, (d) DSPNet, (e) PSRT, (f) MIMO-SST, (g) SMGU-Net, (h) SFIGNet, (i) Ours, and (j) GT.

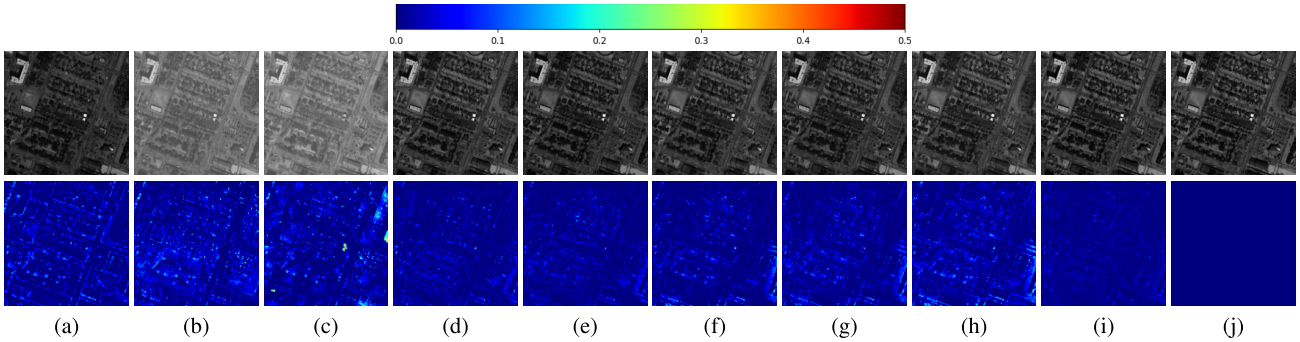


Fig. 13. Single-band grayscale and error maps (26th band) of the fusion results of different algorithms on the Houston dataset. (a) CNMF, (b) NSTF, (c) JSSO, (d) DSPNet, (e) PSRT, (f) MIMO-SST, (g) SMGU-Net, (h) SFIGNet, (i) Ours, and (j) GT.

TABLE II
QUANTITATIVE EVALUATION OF DIFFERENT ALGORITHMS ON HOUSTON DATASET

Methods	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
CNMF	0.4484	5.3639	9.7788	4.3475	37.8793	0.9010	0.9565	0.8894
NSTF	0.5554	6.7202	12.1121	6.4010	36.9428	0.8906	0.9491	0.8974
JSSO	1.0097	6.6784	12.5030	6.3801	36.7583	0.8890	0.9425	0.8896
DSPNet	0.3310	3.8262	7.1037	3.4386	37.0156	0.9482	0.9789	0.9591
PSRT	0.3018	3.4273	6.4543	3.1747	37.9179	0.9546	0.9826	0.9646
MIMO-SST	0.4070	4.6280	8.7642	4.2402	36.3840	0.9374	0.9670	0.9457
SMGU-Net	0.3705	4.1932	7.8766	4.0437	36.1746	0.9405	0.9747	0.9510
SFIGNet	0.4243	4.5795	9.2276	4.1557	38.0789	0.9338	0.9683	0.9436
Ours	**0.2068**	**2.3341**	**4.3969**	**2.4393**	**41.1018**	**0.9704**	**0.9921**	**0.9829**

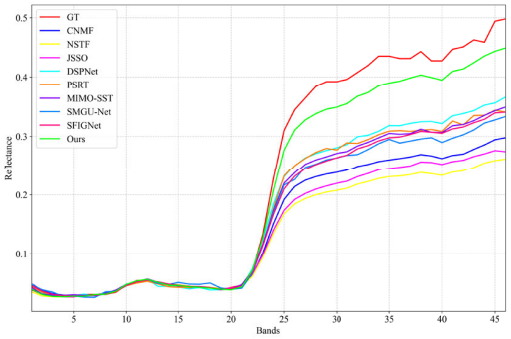
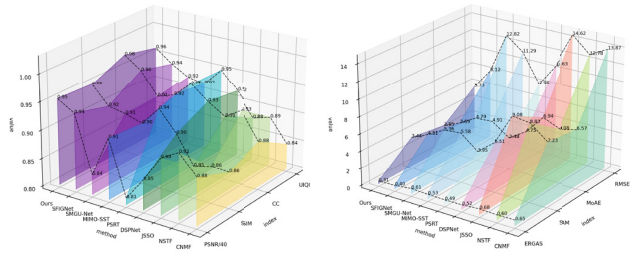


Fig. 14. Spectral curves of fusion results of different algorithms on the Houston dataset.

2) Experimental Results on the Houston Dataset: Table II summarizes the average objective metric values for the fusion results on the Houston dataset. As indicated in Table II, SEMF-Net obtains the optimal values for all metrics. The proposed method achieves a PSNR value that is 3.0229 dB higher than the suboptimal result, alongside a SAM value reduced by 0.7354, demonstrating that SEMF-Net yields fusion outcomes with superior spatial fidelity and minimized spectral distortion. The PSRT obtains suboptimal values for seven metrics, which indicates that the method is also able to perform image fusion well. In addition, all algorithms perform better on the Harvard dataset than the Houston dataset, indicating that different data distributions result in different fusion difficulties, and SEMF-Net can obtain better fusion results with increased fusion difficulty. Similarly, Fig. 11 visualizes the metrics in Table II, from which we intuitively see that the objective metrics of the proposed algorithm outperform the other algorithms.

Fig. 12 visualizes the fusion outcomes of the "2_2" image in the form of pseudo-color images. Evidently, most of the fusion algorithms have serious chromatic aberration at the yellow buildings, while SEMF-Net is not only closer to the GT in color but also has clearer spatial details, which indicates that SEMF-Net can more accurately fuse the spatial–spectral feature of the source images. In order to facilitate the observation of the errors of each fusion algorithm, Fig. 13 displays the fusion grayscale and error maps of the "2_2" image at the 26th band. The grayscale maps of NSTF and JSSO are obviously brighter than that of GT, which suggests that they erroneously enhance the pixel intensity of this band in the fusion process. Overall, the error map of SEMF-Net has the lowest brightness and produces the smallest fusion error. To demonstrate the accuracy of the spectral information, Fig. 14 plots the spectral profiles at random locations of the "2_2" image. The closer to the spectral edge, the more likely it is to produce spectral errors, which indicates that it is meaningful

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(a) (b)

Fig. 15. Visualization of quantitative evaluation of different algorithms on the PaviaU dataset. (a) PSNR, SSIM, CC, and UIQI. (b) ERGAS, SAM, MoAE, and RMSE. Larger colored areas in (a) indicate better metrics, and vice versa in (b).

TABLE III
QUANTITATIVE EVALUATION OF DIFFERENT ALGORITHMS ON THE PAVIAU DATASET

Methods	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
CNMF	0.6453	6.5749	13.8650	7.2306	35.3747	0.8594	0.8813	0.8450
NSTF	0.5976	6.0795	12.7829	8.8332	36.7163	0.8611	0.9291	0.8863
JSSO	0.6778	6.9363	14.6218	9.0827	36.0062	0.8533	0.9091	0.8772
DSPNet	0.5225	4.7502	10.6332	5.5134	34.1199	0.9019	0.9284	0.9188
PSRT	0.4863	3.4810	7.9566	3.9506	32.4820	0.9375	0.9527	0.9501
MIMO-SST	0.5335	4.9050	11.2872	5.5798	36.3691	0.9038	0.9246	0.9176
SMGU-Net	0.6102	4.7879	12.8205	5.3615	33.4182	0.9128	0.9143	0.9225
SFIGNet	0.4026	3.6878	8.1239	4.3118	37.4415	0.9184	0.9554	0.9396
Ours	**0.3077**	**2.8479**	**5.7259**	**3.4380**	**38.1554**	**0.9505**	**0.9763**	**0.9623**

to study the enhancement of spectral edges. Furthermore, the result of SEMF-Net is closer to GT across the entire spectral band, which indicates that the SEME block plays the role of enhancing the spectral edges and realizes a more accurate fitting of spectral information.

3) Experimental Results on the PaviaU Dataset: In this article, we further conducted comparative experiments for each fusion algorithm on the PaviaU dataset. As demonstrated in Table III, the SEMF-Net attains optimal performance across all evaluated metrics. Meanwhile, the waterfall plot of the objective metrics in Fig. 15 visualizes that the fusion performance of the SEMF-Net is superior to other algorithms. Notably, the RMSE value of SEMF-Net is 2.2307 lower than the suboptimal value, which indicates that the fusion results of SEMF-Net do not have a large error and are overall closer to GT. PSRT obtains the suboptimal value in six indicators, which indicates that the algorithm also has a strong multimodal information extraction and fusion capability.

In this article, a test image is randomly extracted from the PaviaU dataset, and the test results of each fusion algorithm are converted into pseudo-color to show in Fig. 16, with a detailed zoomed-in view provided for the region marked by the green box. As evidenced by the magnified view, most of the comparison algorithms show obvious color aberrations in local regions with high luminance, which may be due to the excessive focus on luminance fidelity and neglect of color distribution in the fusion process. In contrast, the SEMF-Net can adjust parameters to adapt to the differentiated local features dynamically and maintain the consistency of brightness and color. To show the fusion error more clearly, Fig. 17 plots the grayscale and error maps of the fused images at the 67th band. Consistent with the results on the Houston data, the grayscale maps of NSTF and JSSO are obviously darker. The error map shows that the fusion results of the proposed method have the smallest error and the most complete retention of spatial details. To demonstrate the degree of fit between the spectral information and the GT, we plot the spectral profiles at random locations for the fusion results of each algorithm in Fig. 18. Again, in the band closer to the edge, the spectral curves of most algorithms are farther away from the GT, while the SEMF-Net can better fit the GT over the full band, which indicates the efficacy of the proposed SEME block.

TABLE IV
QUANTITATIVE EVALUATION OF DIFFERENT ALGORITHMS ON THE WV3 DATASET

Methods	CNMF	DSPNet	PSRT	MIMO -SST	SMGU -Net	SFIGNet	Ours
QNR↑	0.0575	0.9691	0.9711	0.9712	0.9705	0.9589	**0.9720**
$D_\lambda \downarrow$	**0.0172**	0.0272	0.0252	0.0251	0.0258	0.0375	0.0243

D. Experimental Results on the Real Dataset

To verify the fusion ability of SEMF-Net on real datasets, comparative experiments are conducted on the WV3 dataset. Table IV presents the average values of QNR and D_λ, in which SEMF-Net obtains the optimal QNR value and the suboptimal D_λ value, indicating that the proposed algorithm achieves a more comprehensive and balanced integration of spatial and spectral information. Although CNMF obtains the optimal value of D_λ, its QNR value is very low. This is because the fusion results of CNMF incorporate a large amount of high-frequency spatial detail, enabling clear object resolution in its pseudo-color images. However, the excessive contrast leads to a loss of spatial consistency with the source images.

To facilitate a visual comparison of each algorithm on the WV3 dataset, Fig. 19 shows the pseudo-color images of PAN as well as MSI and fusion results. As illustrated in Fig. 19, the fused images of CNMF and PSRT have blurred spatial information, and the fused image of SFIGNet has low overall image saturation. Overall, the fused image of SEMF-Net has more detailed spatial details and more accurate color information.

E. Ablation Experiments

To validate the contribution of each component in SEMF-Net, we perform ablation studies on the Houston dataset. An investigation of the influence of stage quantity on model efficacy is conducted first. Then, the effectiveness of the SEME block, SEE block, DAIF block, spectral and spatial partitioning strategies in the Ptnformer and DSA block, and the GLSA in the DSA block are evaluated.

1) Effect of the Number of Model Stages: To optimize model performance, we evaluate the fusion efficacy of SEMF-Net with the stage set to 1, 2, 3, 4, and 5. Using a single stage as the baseline, the fusion effects of multiple stages were compared. The quantitative results summarized in Table V demonstrate that the model achieves peak performance when configured with four stages. When the number of stages is less

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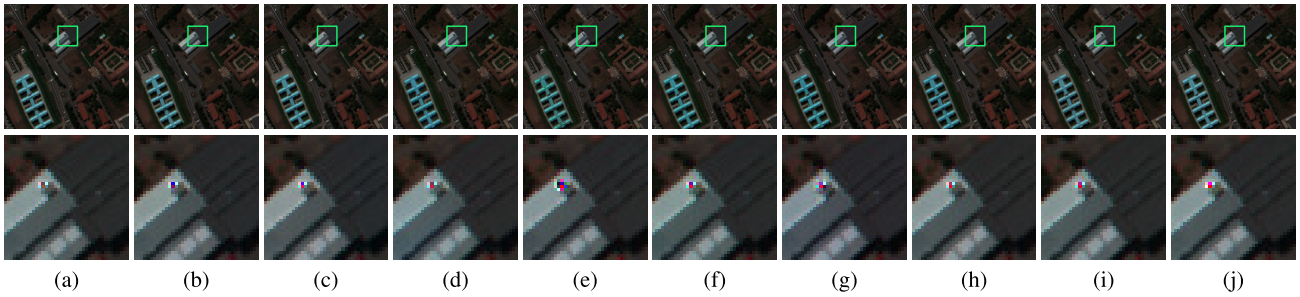


Fig. 16. Pseudo-color images (the 65th, 30th, and 20th spectral bands) of fusion results of different algorithms on the PaviaU dataset. (a) CNMF, (b) NSTF, (c) JSSO, (d) DSPNet, (e) PSRT, (f) MIMO-SST, (g) SMGU-Net, (h) SFIGNet, (i) Ours, and (j) GT.

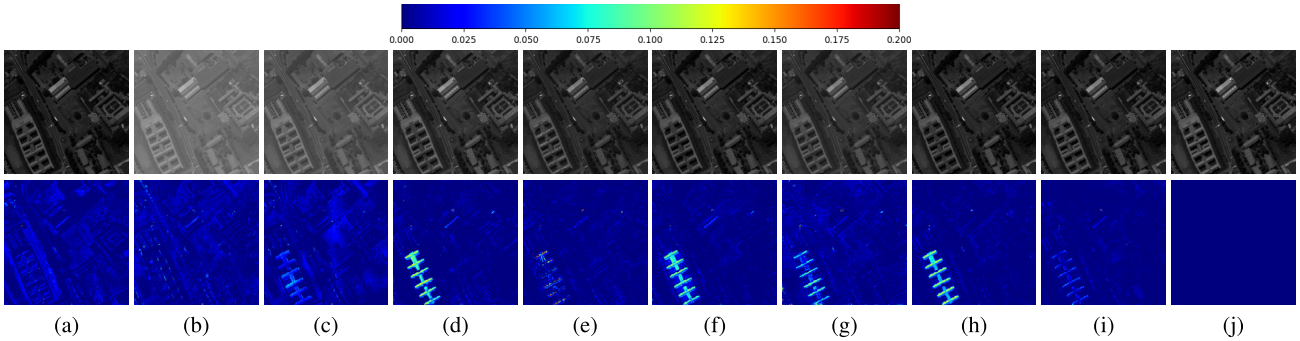


Fig. 17. Single-band grayscale and error maps (67th band) of the fusion results of different algorithms on the PaviaU dataset. (a) CNMF, (b) NSTF, (c) JSSO, (d) DSPNet, (e) PSRT, (f) MIMO-SST, (g) SMGU-Net, (h) SFIGNet, (i) Ours, and (j) GT.

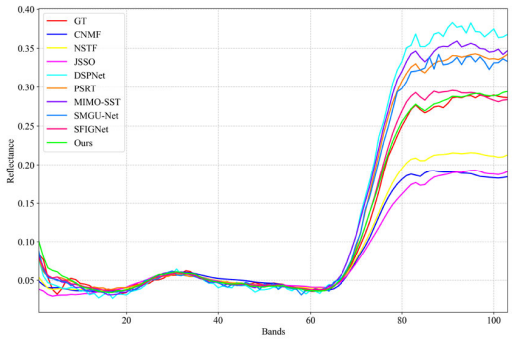


Fig. 18. Spectral curves of fusion results of different algorithms on the PaviaU dataset.

TABLE V

QUANTITATIVE EVALUATION OF THE PROPOSED ALGORITHM WITH DIFFERENT NUMBER OF STAGES

Number of stages	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
1	0.3643	4.1717	7.7280	3.8504	35.6581	0.9425	0.9758	0.9553
2	0.3414	3.9576	7.1444	3.5901	35.6259	0.9470	0.9797	0.9585
3	0.3296	3.8229	7.0672	3.3855	36.9409	0.9503	0.9791	0.9587
4	**0.2068**	**2.3341**	**4.3969**	**2.4393**	**41.1018**	**0.9704**	**0.9921**	**0.9829**
5	0.3165	3.6291	6.8445	3.2048	38.2833	0.9542	0.9806	0.9605

than 4, the performance of the model may be poor due to the limited fitting ability, while when the number of stages is 5, the model may have undergone overfitting and could not reach the optimal level on the test set. Therefore, in this article, the number of stages of SEMF-Net was finally set to 4.

2) Effectiveness of the Proposed Blocks: The SEMF-Net focuses on the enhancement of spatial and spectral edges in the fusion process, thus designing the SEME block and SEE block. To study the influence of the two blocks, the SEME block and the SEE block are removed, and the two ablation models "w/o SEME" and "w/o SEE" are constructed for the fusion experiments, respectively. The objective indices of the ablation models are summarized in Table VI, and the fusion performance of the models decreases after removing the SEME block or SEE block. Among them, the SAM index of the model increases significantly after removing the SEME block, which indicates that the SEME block contributes to the enhancement of spectral information. However, after removing the SEE block, the UIQI index of the model decreases significantly, which indicates that the SEE block helps enhance the overall quality of the fused image. In addition, we visualize the edge features output from the SEE block in the last stage, as shown in Fig. 20, which shows that the edges of trees and roads are outlined in the edge feature maps. The SEMF-Net contributes to the preservation of edge details, thereby enhancing the overall quality of the fused result.

TABLE VI

RESULTS OF ABLATION EXPERIMENTS FOR THE PROPOSED BLOCKS

Ablation models	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
w/o SEME	0.3222	3.6978	6.9049	3.3482	37.9450	0.9522	0.9804	0.9602
w/o SEE	0.3212	3.6189	6.8997	3.1339	38.2127	0.9534	0.9802	0.9598
w/o SpaPB	0.3072	3.5691	6.5929	3.1817	37.8186	0.9540	0.9820	0.9630
w/o SpePB	0.3373	3.9277	7.2728	3.4282	37.0417	0.9517	0.9806	0.9602
w/o GLSA	0.3153	3.6094	6.7877	3.2090	37.9408	0.9530	0.9809	0.9611
w/o DAIF	0.3308	3.6805	7.0376	3.3895	36.8813	0.9506	0.9798	0.9603
SEMF-Net	**0.2068**	**2.3341**	**4.3969**	**2.4393**	**41.1018**	**0.9704**	**0.9921**	**0.9829**

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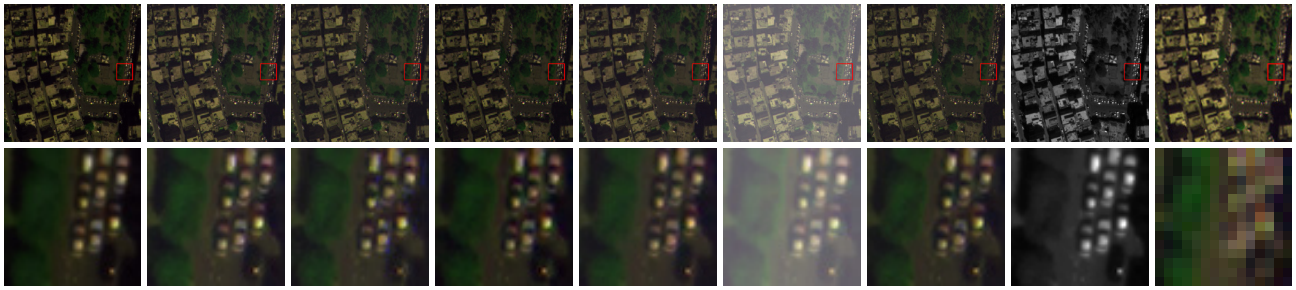


Fig. 19. Pseudo-color images of fusion results of different algorithms on the WV3 dataset. (a) CNMF, (b) DSPNet, (c) PSRT, (d) MIMO-SST, (e) SMGU-Net, (f) SFIGNet, (g) Ours, (h) PAN, and (i) MSI.

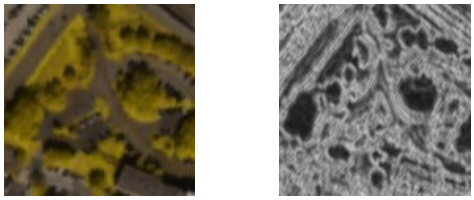


Fig. 20. Visualization of spatial edge feature maps. (a) Reference image and (b) edge feature map.

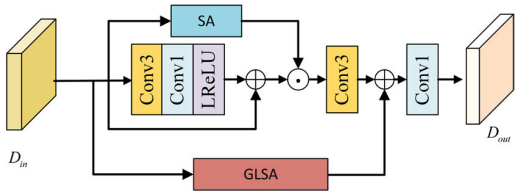


Fig. 21. Network structure of the DSA block without SpaPB.

To verify the validity of SpePB and SpaPB, experimental analysis by removing SpaPB and SpePB is conducted from the DSA block and Ptnformer, respectively. The structure of the DSA block after removing SpaPB (w/o SpaPB) is shown in Fig. 21, and that of the Ptnformer after removing SpePB (w/o SpePB) is the ordinary spectral dimensional Transformer. The objective metrics of "w/o SpaPB" and "w/o SpePB" are presented in Table VI. After removing either SpaPB or SpePB, all the metrics of the model have decreased significantly. Especially after removing SpePB, the RMSE index of the model becomes poor, which indicates that SpePB can reduce the fusion error to a large extent.

The GLSA block is able to perceive spatial features from global and local perspectives, respectively, and the DAIF block is able to realize the inheritance of fused features and perform channel and spatial attention in different feature spaces. To validate the contributions of these two blocks, we replaced the GLSA block with a 3×3 convolutional block in the ablation study, denoted as "w/o GLSA," and replaced the DAIF block with simple feature aggregation (i.e., cascade and convolution), denoted as "w/o DAIF," whose objective metrics are shown in Table VI. After removing both the GLSA block and the DAIF block, the performance of the model decreases significantly, which indicates that it can effectively enhance the fusion performance.

TABLE VII

RESULTS OF ABLATION EXPERIMENTS FOR THE THRESHOLD IN PTNFORMER

Datasets	K	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
Houston	0.4	0.2510	2.8879	5.4268	2.7618	39.9139	0.9649	0.9879	0.9748
	0.5	**0.2068**	**2.3341**	**4.3969**	**2.4393**	**41.1018**	**0.9704**	**0.9921**	**0.9829**
	0.6	0.2523	2.8814	5.4456	2.8344	39.9299	0.9642	0.9878	0.9743
PaviaU	0.4	0.3734	3.2503	7.0475	3.7506	36.1130	0.9421	0.9655	0.9543
	0.5	**0.3077**	**2.8479**	**5.7259**	**3.4380**	**38.1554**	**0.9505**	**0.9763**	**0.9623**
	0.6	0.3725	3.4539	6.4225	3.8721	35.0894	0.9438	0.9717	0.9569

TABLE VIII

RESULTS OF ABLATION EXPERIMENTS FOR k IN THE SEME BLOCK

k	ERGAS	MoAE	RMSE	SAM	PSNR	SSIM	CC	UIQI
3	0.3217	3.7649	6.9115	3.3397	37.4973	0.9524	0.9806	0.9606
4	0.2497	2.9066	5.3960	2.8360	39.9396	0.9643	0.9879	0.9744
5	**0.2068**	**2.3341**	**4.3969**	**2.4393**	**41.1018**	**0.9704**	**0.9921**	**0.9829**
6	0.2591	2.9750	5.5938	2.8386	39.7273	0.9630	0.9871	0.9727

3) Ablation Study of the Threshold in Ptnformer: For the partitioning threshold K in Ptnformer, we conducted ablation studies on two datasets. In particular, experiments were performed on the Houston and PaviaU datasets with K set to 0.4, 0.5, and 0.6, respectively. The objective metrics are shown in Table VII. Table VII demonstrates that the optimal results were achieved when K was set to 0.5, regardless of whether it was the Houston dataset with fewer bands or the PaviaU dataset with more bands. Therefore, we set the K to 0.5.

4) Ablation Study of k in the SEME Block: The SEME block enhances spectral information by mirroring and concatenating spectral edges. An ablation study was conducted for the number of flipped bands k. The experiments were performed on the Houston dataset with k set to 3, 4, 5, and 6, yielding objective metrics as shown in Table VIII. The results indicate that the optimal performance is achieved when k is set to 5. Furthermore, since the SEME block is primarily designed for spectral edges, the choice of k is independent of the total number of bands in the HSI data. Therefore, k is uniformly set to 5 across all datasets in this study.

V. CONCLUSION

In this article, a multistage fusion network based on spatial and spectral edge enhancement, SEMF-Net, is proposed. SEMF-Net consists of an SEME block for spectral edge enhancement, SEE blocks for SEE, and multistage fusion.

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The SEME block and the SEE block can improve spatial and spectral edge features in the model during fusion to minimize the edge loss. In multistage fusion, the Ptnformer can perform long-distance dependencies on spectral subregions, reducing interference from unfavorable information. The DSA block is used for dual-spatial perception: global-local spatial perception on the one hand and subregional spatial perception on the other, enabling more complete modeling of spatial details. The DAIF block realizes the inheritance of the fusion features of the historical stages and the effective fusion of the spatial–spectral information of the current stage. Finally, we experimentally verify that the SEMF-Net can mitigate the loss of spatial–spectral edges as well as the spatial texture of the fusion results and outperforms the existing state-of-the-art algorithms both subjectively and objectively. However, this article directly up-samples the LR-HSI to the target size and lacks the exploration of features at different spatial scales, which we will address and improve in the model in the future.

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